October 12, 2026

Dear Prof. Karl von Ellenrieder, Editor-in-Chief, and the Editorial Board of the IEEE Journal of Oceanic Engineering,

We are pleased to submit the enclosed manuscript, "Honest Evaluation of Underwater Acoustic Target Recognition: Decomposing, Predicting, and Reducing Benchmark Variance," for consideration as a Research Article in the IEEE Journal of Oceanic Engineering.

The community guidance in underwater acoustic target recognition (UATR) increasingly requires recording- or vessel-disjoint evaluation, but the statistical cost of this convention has never been quantified. Our manuscript supplies the missing half: on OceanShip, identity-disjoint evaluation multiplies draw-to-draw noise twelve-fold over clip-level evaluation; we show that 89% of this variance is explained by the class composition of the test set (59% predictable out-of-sample under leave-one-draw-out); we validate a first-order composition predictor across five allocation schemes and three model families; and we show that a class-wise clip-mass stratified allocation reduces evaluation variance 14.4-fold (exact permutation $p = 0.004$), cutting the draws needed to resolve a 2 pp improvement from about 100 to 8 per system. A falsifiable negative control on DeepShip, and a six-part diagnosis of the remaining tail-class collapse, bound the claims.

We believe this work fits the Journal's scope of original, analytical contributions to ocean science and technology: it does not propose a new classifier, but instead quantifies, predicts, and controls the uncertainty that identity-disjoint benchmarks introduce—turning "a stricter convention" into "a protocol with known, managed statistics."

Declarations accompanying this submission: (i) the manuscript is original, has not been published previously, and is not under consideration elsewhere in any language; (ii) all frozen splits, per-draw results, training and aggregation code, and the run manifest are publicly released at `https://github.com/gzhgzy123456` in connection with this submission; (iii) the authors declare no competing interests, financial or otherwise; (iv) this research involved no human subjects and no animals; (v) the submission is made with the full knowledge and consent of all co-authors, and the corresponding author will keep them informed of all correspondence.

We suggest no specific reviewers, but would be glad to point to colleagues working on evaluation methodology if the Editor wishes. Thank you for considering our manuscript.

Sincerely,

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